

PRATIK PATIL

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience across the full development lifecycle: designing REST APIs, building and deploying production-style applications, working with relational databases, and automating delivery through CI/CD pipelines. Proficient in Python, SQL, Git, Docker, and cloud platforms (AWS, GCP, Azure), with additional specialization in machine learning and generative AI. Comfortable working independently or in cross-functional teams to ship reliable, well-documented, maintainable software.

TECHNICAL SKILLS

- Languages & Core CS: Python, SQL, Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Development Lifecycle (SDLC), Debugging, Unit Testing (PyTest)
- Backend & APIs: FastAPI, REST API Design & Development, Microservices-style Architecture, Agentic Workflows
- DevOps & Deployment: Docker, Git, GitHub Actions (CI/CD), Version Control, Cloud Platforms (AWS, GCP, Azure), Application Monitoring & Logging
- Data & Databases: SQL, Excel, Power BI, Pandas, NumPy, Data Visualization (Matplotlib, Seaborn)
- AI/ML Specialization: Machine Learning, Deep Learning (TensorFlow, PyTorch), Generative AI, Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LangChain, Computer Vision (OpenCV)
- Ways of Working: Agile collaboration, Technical Documentation, Cross-functional Communication, Stakeholder Reporting

PROFESSIONAL EXPERIENCE

Warehouse Systems Operator — Frasers Group (Distribution Operations)

Oct 2025 – Present

- Operated Zebra MC3300x enterprise mobile computers to process high-volume deliveries through a Warehouse Management System (WMS), scanning delivery containers to initiate records and individual items to validate and update inventory in real time.
- Maintained inventory data accuracy across a large-scale automated distribution operation by reconciling expected vs. scanned quantities, catching discrepancies before they propagated downstream to stock and fulfillment records.
- Handled exception cases (failed scans, quantity mismatches, damaged stock) through a structured troubleshoot-recheck-escalate process, coordinating with team leads to resolve unresolved issues.
- Built Excel/SQL/Power BI dashboards and automated compliance and training delivery across SuccessFactors and Workday Learning, consolidating fragmented reporting and cutting onboarding processing time by 12%.
- Supported a 10-15 member team processing up to 15 deliveries per day at 100+ units per delivery, contributing to accurate, real-time synchronization between physical stock and the digital inventory system.

Global Shala — Project Intern (Virtual, USA, via Saint Louis University)

Feb 2022 – Mar 2022

- Accomplished successful delivery of the "Super Hero YOU" initiative, as measured by on-time completion across a multicultural virtual team, by leading project planning, cross-functional coordination, and stakeholder presentations.
- Strengthened leadership, communication, and problem-solving skills directly applicable to software delivery teams.

SOFTWARE & DATA PROJECTS

Production ML Inference API | FastAPI, Docker, GitHub Actions, PyTest, MLOps

GitHub: <https://github.com/MrPratik05/production-ml-inference-api>

- Accomplished a production-style REST API service, as measured by reliable automated request handling, by building a FastAPI backend integrated with Hugging Face Transformers.
- Accomplished consistent, reliable releases, as measured by automated build/test/deploy cycles, by implementing CI/CD pipelines with GitHub Actions and automated testing via PyTest.

- Accomplished improved system observability, as measured by faster issue detection, by adding monitoring, health checks, and structured logging.
- Accomplished reproducible deployments, as measured by consistent runtime behavior across environments, by containerizing the application with Docker.

AssistFlow AI — Agentic Customer Support Platform / *Python, LangChain, LangGraph, Docker, GitHub Actions*

GitHub: <https://github.com/MrPratik05/assistflow-ai>

Live Demo: <https://huggingface.co/spaces/MrPratik05/assistflow-ai>

- Accomplished a modular, agentic backend application, as measured by successful intent classification, retrieval, and escalation handling, by architecting workflows in LangChain and LangGraph.
- Accomplished a deployed, production-style application, as measured by live public availability, by building a Streamlit interface, containerizing with Docker, deploying to Hugging Face Spaces, and integrating GitHub Actions CI.
- Accomplished secure, maintainable architecture, as measured by adherence to modular design standards, by implementing secure API key management and comprehensive project documentation.

AI Financial Analysis Chatbot / *Python, Streamlit, Pandas, GitHub*

GitHub: <https://github.com/MrPratik05/financial-analysis-chatbot>

Live Demo: <https://financial-analysis-chatbot-p9lk5jtkda8mdqvydoyi3h.streamlit.app/>

- Accomplished a natural-language query interface over structured financial data, as measured by accurate responses across revenue, cash flow, and balance sheet queries, by building a Python and Pandas-based query engine.
- Accomplished a live, publicly accessible tool, as measured by successful deployment, by designing an interactive Streamlit front end integrated with GitHub-based version control.

Quantitative Research — Industry Simulation / *Python, Statistical Modeling — JPMorgan Chase & Co. x Forage*

- Accomplished natural gas price forecasting, as measured by trend-aligned projections, by building a model using historical market data, interpolation, and trend analysis.
- Accomplished credit risk estimation, as measured by Probability of Default (PD) outputs, by implementing a Logistic Regression model on borrower financial characteristics.
- Accomplished a borrower risk segmentation framework, as measured by consistent score quantization, by designing a FICO-based rating system.

AI-Powered Delinquency Prediction & Collections Strategy / *Python, Decision Trees — Tata Group x Forage*

- Accomplished a customer delinquency risk model, as measured by classification accuracy, by developing a Decision Tree-based predictive analytics framework.
- Accomplished data-driven collections recommendations, as measured by stakeholder-ready strategy proposals, by designing collections strategies grounded in risk segmentation and EDA insights.

Animal Image Classification — Computer Vision / *Python, OpenCV, TensorFlow*

- Accomplished a robust image classification pipeline, as measured by clean, balanced training data, by using OpenCV for preprocessing, augmentation, and feature extraction.
- Accomplished optimized model performance, as measured by interpretation of confidence scores and outputs, by training and evaluating deep learning classification models.

Crop Yield Prediction Using Machine Learning / *Python, Random Forest, Gradient Boosting*

- Accomplished high-accuracy yield forecasting, as measured by model performance across historical yield and weather data, by developing and benchmarking Random Forest, Gradient Boosting, and Neural Network models.

Transformer-Based Text Summarization Model / *Python, TensorFlow/PyTorch, NLP*

- Accomplished automated text summarization, as measured by improved summary quality, by implementing generative AI and sequence-to-sequence NLP techniques.

µGenAI: Lightweight Generative AI Models / *MSc Project — TensorFlow, PyTorch*

- Accomplished optimized generative AI models for embedded systems, as measured by reduced model size and footprint, by applying quantization, pruning, and knowledge distillation.

Spiking Neural Network for Audio Processing / *NEST*

- Accomplished audio-to-spike and spike-to-audio conversion, as measured by functional encode-decode performance, by designing a Spiking Neural Network (SNN) in NEST.

EDUCATION

- MSc Artificial Intelligence — Nottingham Trent University, UKOct 2025
- B.Eng. Mechanical Engineering — Sandip Institute of Technology and Research Centre, IndiaJul 2024
 - Project: Smart Solar Drier — designed and developed an energy-efficient solar drying system (patent filed).

CERTIFICATIONS

- Introduction to Retrieval Augmented Generation (RAG) — Coursera (Duke University), Apr 2026 | Credential ID: MVOFCAOMOKU3
- Generative AI with Large Language Models — Coursera (DeepLearning.AI & AWS), Apr 2026 | Credential ID: JDKM2NP85MG3
- MCP — Model Content Protocol — Coursera, May 2026 | Credential ID: 82UBVOLJ9L49
- Quantitative Research Job Simulation — JPMorgan Chase & Co. x Forage, Jun 2026 | Credential ID: ZDfxdomj4YYNARgab
- GenAI Powered Data Analytics Job Simulation — Tata Group x Forage, Jun 2026 | Credential ID: TJBDXAPZsDMBg4NZB
- Microsoft Azure AI Fundamentals (AI-900) — Microsoft, Sep 2025 | Credential ID: Wnm8J-2FMo

ACHIEVEMENTS

- 2nd Place, Project Competition — SPPU, Pune & Institute of Engineers (India), 2022
- Runner-up, Wrestling — Zonal Sports Meet, 2018

LANGUAGES

English (Fluent), Hindi (Proficient), Marathi (Native), German (Basic)

ADDITIONAL INFORMATION

- Volunteer: Mega Job Fair; National Service Scheme (NSS)
- Member: Mechanical Engineering Student Association (MESA); Rotaract Club of Nashik Nine Hills; The Institution of Engineers (India)